

Preprocessing Steps & Business Questions

India's Agricultural Crop Production Analysis — Project Development Phase

Project Title	India's Agricultural Crop Production Analysis
Domain	Agriculture / Data Analytics & Business Intelligence
Tool	Tableau Desktop / Tableau Public
Dataset	Historical Indian Crop Production Data, 1997–2021
Prepared By	Divya Kauluri (Team Project — completed individually; the team lead and other assigned members did not contribute to the work)
Tableau Public Link	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject
GitHub Repository	https://github.com/divyakauluri-spec/India-s-agricultural-crop-production-analysis
Date	02/07/2026

1. Data Preprocessing Steps

Before any chart was built, the raw government crop production extract was cleaned and enriched using Python 3.12 with the Pandas library. The table below documents every preprocessing step in the order it was applied.

Step	Action	Why It Was Needed
1. Load Raw Extract	Read the government crop production CSV (345,407 rows, 1997–2021) into a Pandas DataFrame.	The source file arrives as a single flat file with no relational structure, so every subsequent check has to be built manually.
2. Remove Duplicates & Blanks	Drop fully duplicated rows and rows with blank/null State, Crop, or Year values.	Duplicate or blank rows would double-count or silently break aggregations like SUM(Production) in every chart.
3. Standardise Names	Normalise spelling variants in State_Name and Crop (e.g., trimming whitespace, fixing casing).	Inconsistent spelling would split one state into multiple map regions on the State-wise Production Map.
4. Derive Crop_Year_Numeric	Convert the text-based Crop_Year field (e.g., '2008-09') into a clean integer year.	A numeric year is required to drive a continuous time axis on the Trend of Agricultural Yield and Year-wise Production charts.
5. Compute Yield	Add a calculated field: $\text{Yield} = \text{Production} \div \text{Area}$.	Yield does not exist in the raw extract but is essential for measuring productivity independent of farm size.
6. Compute Production_Normalized	Add a calculated field converting Production to a common tonnage-equivalent basis where feasible.	Production_Units differ by crop (Tonnes, Bales, Nuts); without normalisation, raw totals conflate incompatible units.
7. Convert to Millions	Add Area (M) and Production (M) fields, dividing Area and Production by 1,000,000.	Raw totals run into the hundreds of millions/billions; scaling to millions keeps chart axes and

		KPI cards readable.
8. Export Cleaned CSV	Write the cleaned, enriched DataFrame to skillwallet_crop_cleaned.csv.	This is the single file Tableau connects to, keeping the visualization layer decoupled from the raw government format.

Data Quality Checks Applied

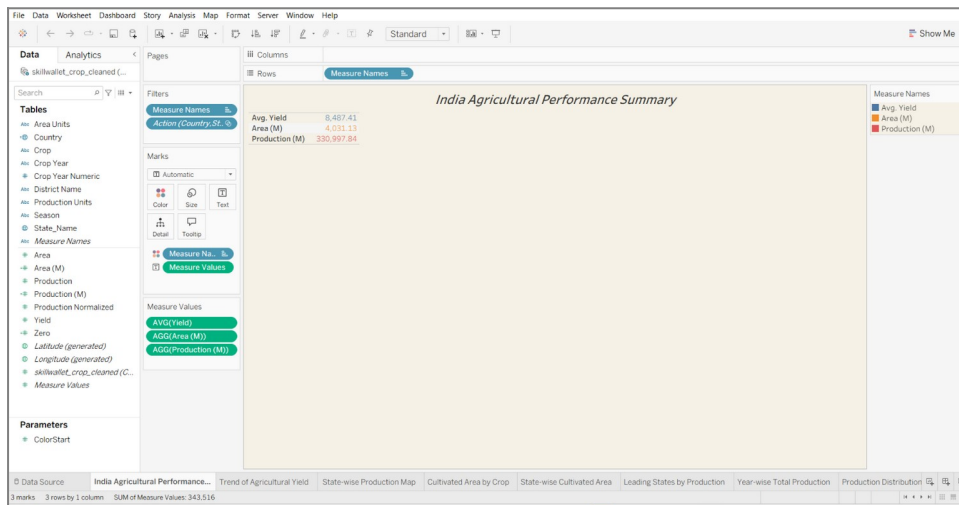
- Row-count reconciliation: the cleaned file's row count was compared against the raw file's row count after removing duplicates/blanks, to confirm no unintended data loss.
- Range checks: Area and Production values were checked for negative or zero entries that would indicate a data entry error rather than a genuine zero-production season.
- Unit consistency: Production_Units was cross-checked against Crop to confirm, for example, that Coconut is consistently recorded in Nuts and Cotton in Bales, rather than mixed inconsistently within the same crop.
- Partial-year flag: 2021 records were flagged as a partial-year data limitation, since year-end reporting was still incomplete at the time of data collection — visible as the drop at the end of the Trend of Agricultural Yield line chart.

2. Business Questions & Visualizations

Each core business question below was defined during Requirement Analysis (Section 2.4, User Stories) and answered by one or more of the nine Tableau worksheets. The screenshots show the actual worksheet used to answer each question.

Q1 (National Overview): What is India's overall agricultural performance at a glance?

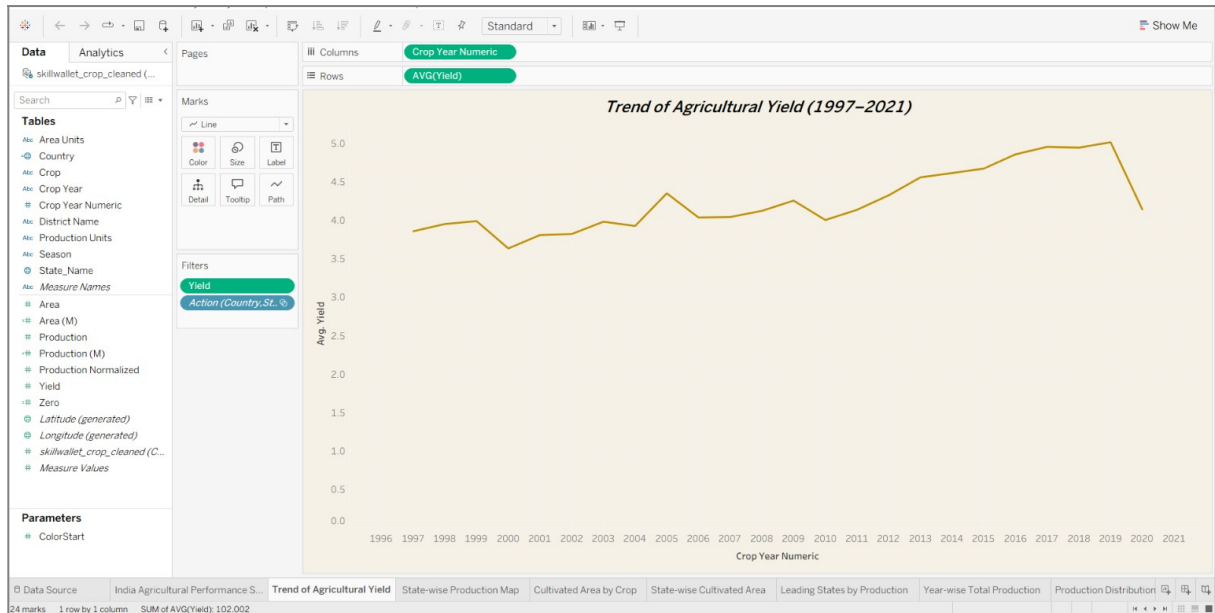
Answered by the KPI Summary panel, which reports Average Yield (8,487.41), total cultivated Area (4,031.13 M hectares), and total Production (330,997.84 M) across the full 1997–2021 dataset — giving any viewer, regardless of role, an instant national baseline before drilling into detail.



Q1 (National Overview)

Q2 (Trend, Policy Analyst): Has national crop yield improved or declined over the past 25 years?

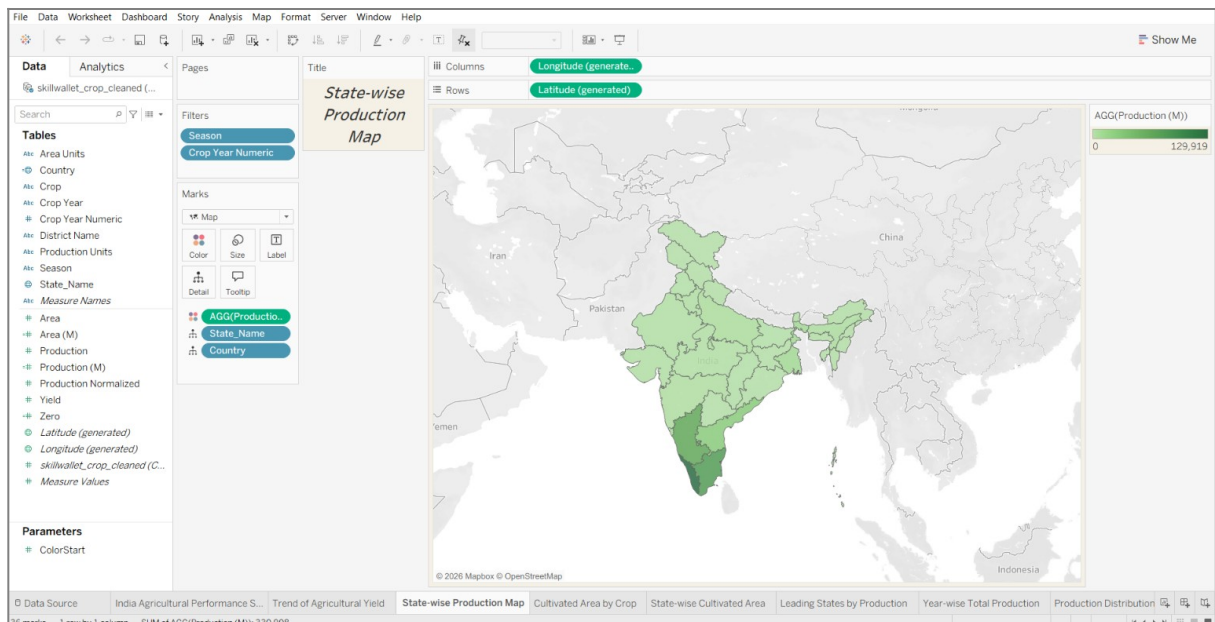
Answered by the Trend of Agricultural Yield line chart. Yield rises from just under 4.0 in 1997 to a peak above 4.75 around 2017–2019, confirming steady long-term productivity gains, with the final year's dip attributable to incomplete 2021 reporting rather than an actual decline.



Q2 (Trend, Policy Analyst)

Q3 (Policy Analyst): Which states or regions produce the most, and where should investment focus?

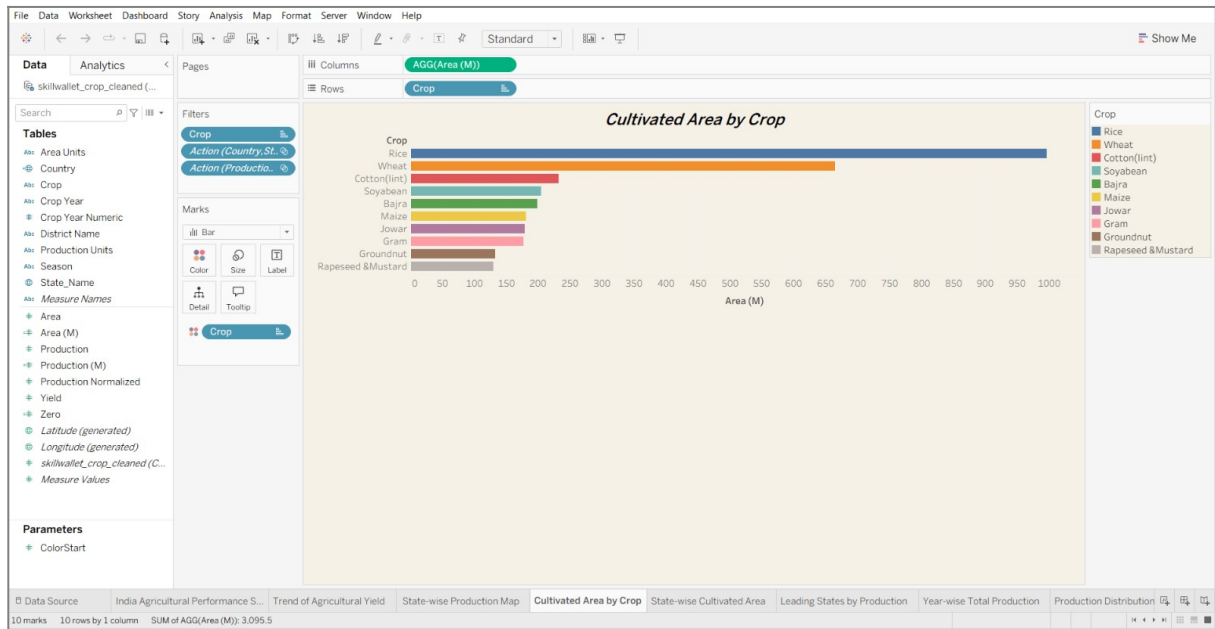
Answered by the State-wise Production Map, which shades every state by total production so regional disparities are visible at a glance without reading a 36-row table.



Q3 (Policy Analyst)

Q4 (Farmer): Which crops occupy the most cultivated land nationally, and how does that compare to what I'm growing?

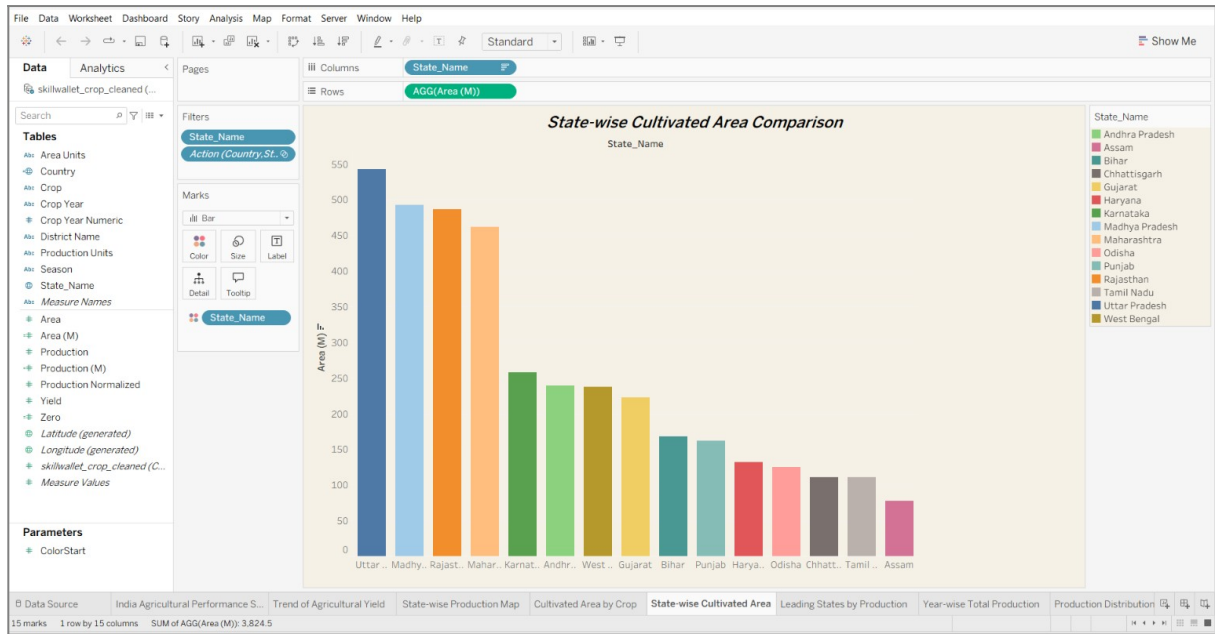
Answered by the Cultivated Area by Crop bar chart. Rice and Wheat dominate national land use, followed by Cotton(lint) and Soyabean, giving a farmer a benchmark against his own crop choice.



Q4 (Farmer)

Q5 (Investor): Which states have the largest agricultural land base to support scaled production?

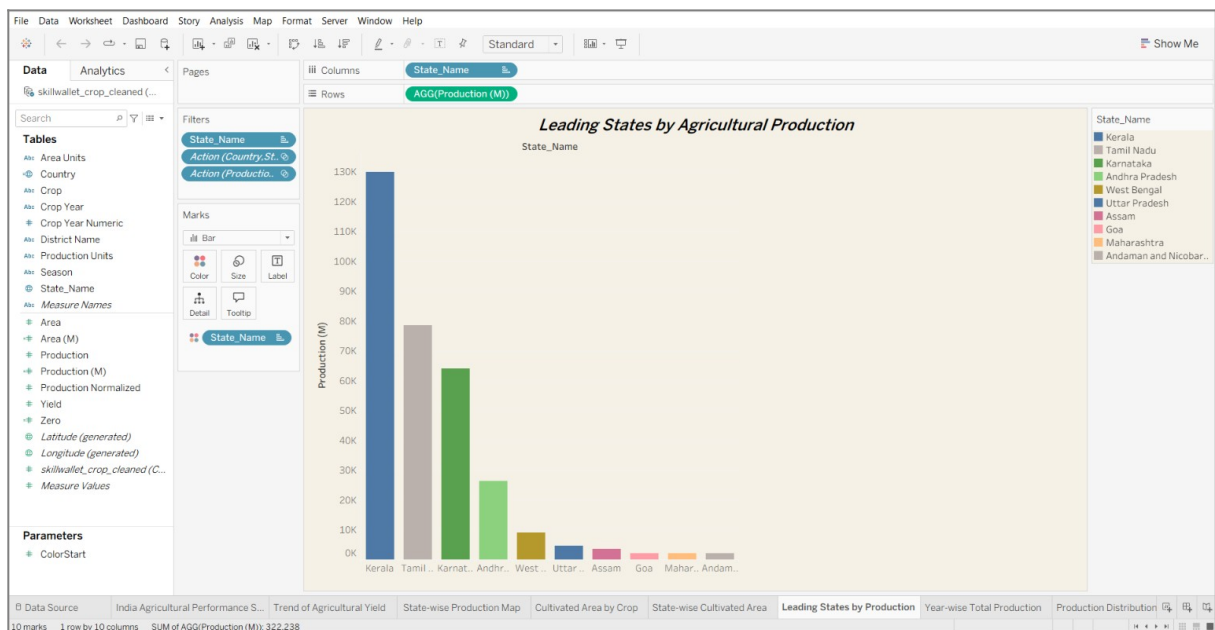
Answered by the State-wise Cultivated Area Comparison chart. Uttar Pradesh, Madhya Pradesh, Rajasthan, and Maharashtra lead by a wide margin.



Q5 (Investor)

Q6 (Investor): Which states generate the highest production volume, and is that volume reliable for investment?

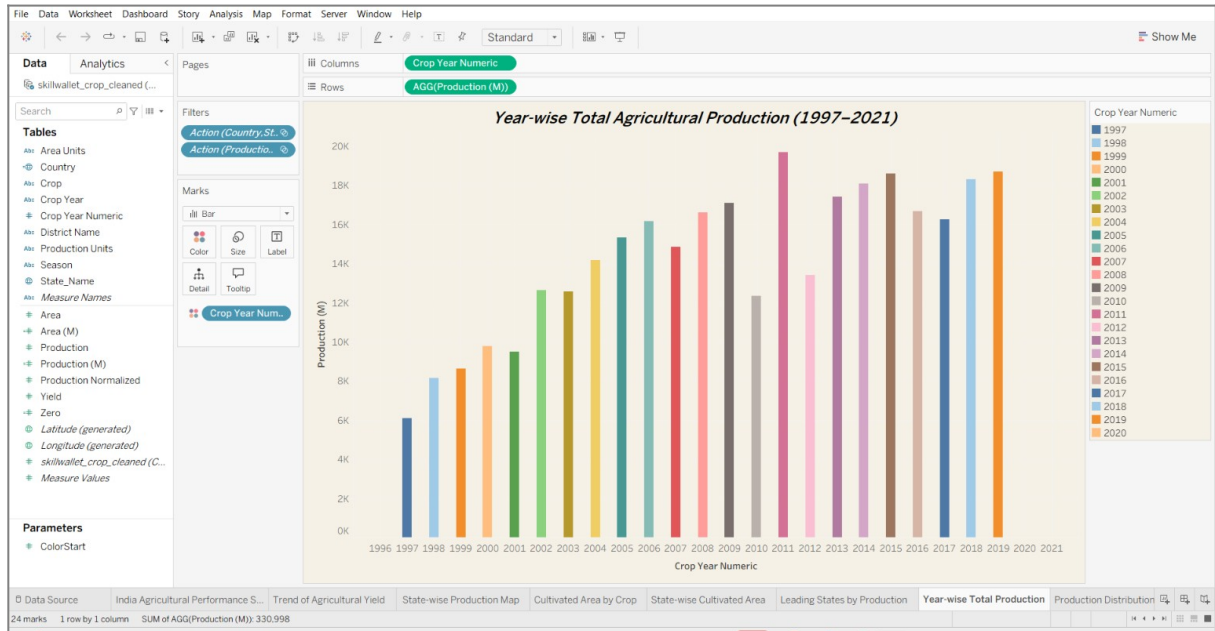
Answered by the Leading States by Agricultural Production bar chart, read alongside Q8 below to correctly interpret the unit basis behind each state's ranking.



Q6 (Investor)

Q7 (Policy Analyst): How has total national production grown year over year, and where did growth plateau?

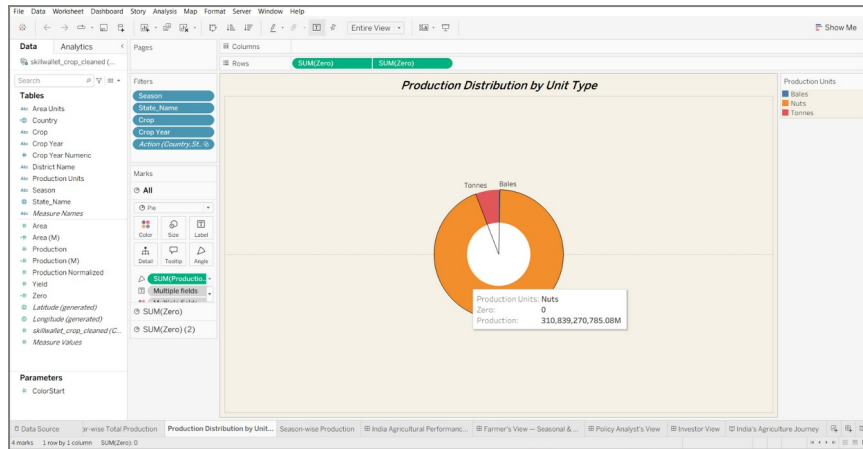
Answered by the Year-wise Total Agricultural Production bar chart, which shows steady growth through the early 2010s followed by fluctuation rather than continued acceleration.



Q7 (Policy Analyst)

Q8 (Investor / Analyst): Are production rankings distorted by mixing incompatible measurement units?

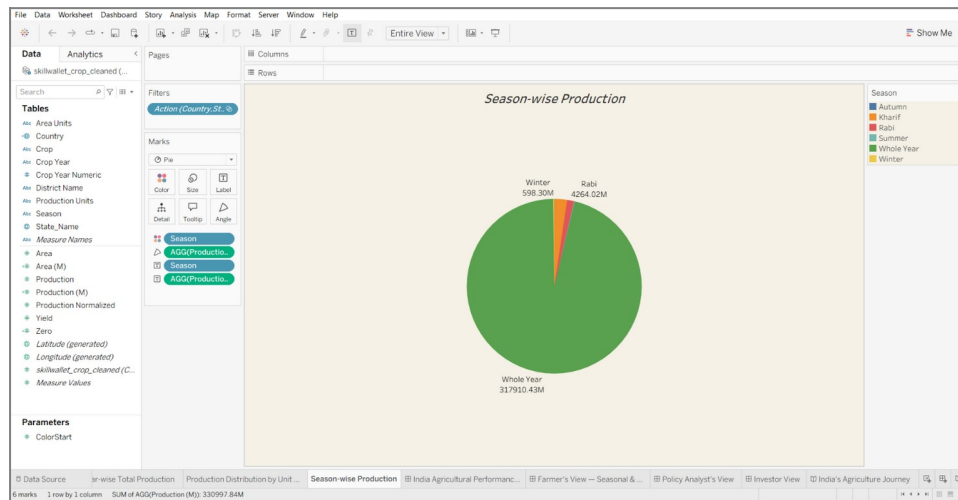
Answered by the Production Distribution by Unit Type donut chart, which reveals that the majority of raw production figures come from crops measured in Nuts (Coconut) — explaining why Kerala and other coconut-producing states rank so highly in Q6, and why that ranking must be read with this unit context rather than taken at face value.



Q8 (Investor / Analyst)

Q9 (Farmer): Which growing season should I prioritise for the crop I want to plant?

Answered by the Season-wise Production pie chart. Whole Year crops dominate national production (317,910.43 M), followed by Rabi (4,264.02 M) and Winter (598.30 M), consistent with India's cropping calendar.



Q9 (Farmer)